

Todo List:

- Shervin
 - Analyse standard GNN interpretability techniques as baseline to SAEs (**priority**)
 - Point 8 in stanford review suggestions
 - Find how to use motif benchmarks (BA-shapes / Tree-Cycles) mentioned in point 8 of Stanford suggestion (an interpretability analysis combining both motif benchmarks and standard GNN interpretability is also there) → (in case time is there)
 - It can connect with Phase 3 of Gemini suggestions
- Manhar
 - Implement and train SAEs (Gated, Switch, JumpReLU, Top-K)
 - Native GNN ablation comparison
 - Get MeanBaseline to work (if time is there)
- Harry
 - **Redo the GAT work with three layers possibly with new SAEs (priority)**
 - Find a suitable real dataset (some suggestions in Point 6 of Stanford Agent Reviewer) and use that to train our GNNs on – (**priority**)
 - Implement and train different architectures (if feasible and time left)
 - Point 8,9 on Stanford Reviewer suggestions

Overall summary of things to add:

Benchmark, **GAT (priority)**, **real data (priority)**, **standard explainability (priority)**, **SAEs and native ablations (priority)**, architectures

Immediate Next Steps:

- Use Gated SAEs and other variants – see how they compare → Manhar
- Finish GATs → Harry
- Understand and analyze standard GNN interpretability techniques

Upcoming Conferences and Deadlines:

<https://github.com/khairulislam/ML-conferences>

Major conferences for us:

1. **ICML – AI/ML (General) – Jan 23 (Main conference)**
2. **IJCAI – AI/ML (General) – Jan 12 (Abstract Submission) – Jan 19 (Full paper for main conference)**
3. **ACL workshop – Linguistics (Need to add justification of why)**

Gemini Next Steps:

Phase 1: The "Real World" Upgrade (Critical for Acceptance)

Goal: Prove that SAEs work on non-synthetic data. **Target Dataset:** **Mutagenicity** (4,337 graphs) or **ZINC** (250k graphs, use a subset). **Why:** These datasets contain known chemical motifs (functional groups like rings, NO2 groups) that determine graph labels (mutagen/non-mutagen).

1. **Train GNN (GCN & GAT):** Train a standard GNN to roughly 80-90% accuracy on Mutagenicity.
2. **Train SAE:** Train your SAE on the hidden layers of this "real" GNN.
3. **Chemical Annotation:**
 - Extract the top 20 most active nodes for a specific SAE feature.
 - Visualize these subgraphs.
 - **The "Money Shot":** Do you see a "Benzene Ring Feature"? A "Nitro Group Feature"?
 - *Note:* You don't need perfect chemistry knowledge; just visual consistency of the subgraph is enough to claim "motif discovery."

Phase 2: Solving the GAT Mystery

Goal: Explain *why* GATs are less interpretable, or fix it. **Hypothesis:** GATs use attention heads to route information, so the "features" might be in the *attention pattern*, not just the node magnitude.

1. **Gated SAEs:** Switch your SAE architecture to **Gated SAEs (Gated ReLU)**.
 - *Reason:* Standard TopK or L1 SAEs struggle with "feature splitting." Gated SAEs are currently SOTA (DeepMind, 2024) and handle polysemanticity better.
 - The SAE encoder is a standard linear–nonlinear Top-K variant; recent work highlights **amortization gaps and recovery limits for such encoders**, which may undermine conclusions about monosemanticity if features are under-recovered.
 - Recent results on SAE encoder insufficiency (2411.13117) suggest that **standard Top-K SAEs can miss features and reduce interpretability fidelity**; this paper should at least acknowledge these limits and, ideally, add experiments with stronger encoders or SAE+inference-time optimization.
 - Given known limitations of linear–nonlinear SAE encoders (e.g., amortization gap and sparse recovery limits), consider **stronger encoders (MLP-encoders or inference-time optimization)** and show that key motif-aligned features and ablation effects persist.

- Recent SAE advances (e.g., **Switch SAEs**, **JumpReLU**, **inference-time optimization**, **encoder expressivity limits**) are not engaged to contextualize the chosen SAE design or its limitations.
 - Scaling and efficiency **advances such as Switch SAEs (2410.08201)** and **best practices in large-scale Top-K training (Llama Scope, 2410.20526)** could inform **better SAE training, regularization, and diagnostic protocols** (e.g., decoder normalization, thresholding schedules, feature utilization analyses).
2. **Attention Head Probe:**
- Instead of just **Activations**, train an SAE on the **Attention Weights** (Edge x Heads).
 - Do specific SAE features correspond to "attending to neighbors in a cycle"?

Phase 3: Baselines & Metrics

Goal: Show that SAEs are *better* than existing tools. **Baseline:** GNNExplainer.

1. **Experiment:**
 - Take a graph with a known motif (e.g., a Feedback Loop).
 - Run GNNExplainer to get a "saliency map".
 - Take your "Feedback Loop SAE Feature" and visualize its activations.
2. **Comparison:**
 - Which one is less noisy?
 - Which one requires less computation at inference time? (SAEs are fast/linear; GNNExplainer requires optimization per example).

Phase 4: Literature & Writing

Narrative Arc for the Paper:

1. **Problem:** GNNs are black boxes. Current tools (GNNExplainer) are "local" (explain one instance) and slow.
2. **Solution:** We need "global" interpretability. We adapt SAEs (successful in LLMs) to GNNs.
3. **Discovery 1:** GCNs naturally learn "Graph Motifs" as monosemantic features (Synthetic + Real World proof).
4. **Discovery 2 (The GAT Insight):** GATs rely less on fixed features and more on [whatever your GAT analysis finds, e.g., "dynamic routing"], presenting a challenge for static dictionary learning.
5. **Impact:** We can now "debug" GNNs by inspecting their feature dictionary.

Suggested Reading List

- **Anthropic:** "Towards Monosemanticity" (The foundation).
- **Google DeepMind:** "Gated Sparse Autoencoders" (The architecture upgrade).
- **NeurIPS:** "GNNEExplainer" (The baseline you must beat).

Stanford Agent Reviewer Suggestions:

Further SAE Statistical Analysis:

1. What are the **distributions** of significant feature–motif correlations ($|r_{pb}|$), per **architecture and per motif, after FDR correction**? How many **features per model are significant**, and how **stable are these features across SAE seeds**?
2. Can you report quantitative **ablation results conditioned on motif presence vs. absence** (e.g., Δ MSE distributions and Wilcoxon p-values per motif), and include **effect sizes** to substantiate selective degradation?
3. How **sensitive** are your conclusions to the SAE design (Top-K vs. L1 sparsity, latent dimensionality L, and K)? Do you observe a **sparsity–interpretability trade-off** akin to SAE studies in LLMs?
 - a. The choice of Top-K sparsification is sensible for monosemanticity but **analyzing sparsity–interpretability trade-offs (varying K; selectivity indices) and feature stability across SAE seeds would better support the “monosemantic” characterization.**
4. main claims of interpretability, where evidence is mostly statistical correlations and a limited ablation. Reporting the following would be helpful:
 - a. counts of **significant feature–motif pairs** per model,
 - b. **distributions of $|r_{pb}|$** across features
 - c. representative feature visualizations (e.g., which nodes/graphs most activate a feature)
 - d. **Ablation effect sizes** conditioned on motif presence vs. absence would substantively improve support.
 - e. **Feature redundancy/duplication analysis (cosine similarity among decoder columns), and dictionary stability across seeds.**
5. Can you report the number/fraction of **significant feature–motif pairs, their effect sizes, and their stability (across seeds/SAE inits) for both GCN and GAT**? This would substantiate the “architecture-dependent interpretability” claim.
 - a. Model selection by maximizing the largest significant correlation across many features risks selection bias; **nested validation or reporting stability across seeds/configs is needed.**

6. The causal ablation operates in the SAE latent space rather than ablation of the actual GNN units; **SAE reconstruction error and encoder/decoder linearity may confound attribution of causal necessity.**
 - a. How **faithful is the SAE reconstruction to the original activations (explained variance, reconstruction MSE, delta task loss)**? Can you also try direct ablations in the original hidden space as a complementary intervention?
 - b. If the SAE does not perfectly reconstruct or linearly embed directions corresponding to true causal features, then ablating a latent dimension may not isolate the intended mechanism. **A complementary ablation in the native GNN space (e.g., lesioning neurons or directions aligned via linear probes) would strengthen causal claims.**
 - c. Faithfulness hinges on the SAE being an accurate, disentangled reparameterization of the true hidden space; **without direct interventions in the GNN representation (e.g., zeroing specific original dimensions or performing activation patching), the conclusion about necessity remains tentative.**
7. Scaling and efficiency advances such as Switch SAEs (2410.08201) and best practices in large-scale Top-K training (Llama Scope, 2410.20526) could inform better SAE training, regularization, and diagnostic protocols (e.g., decoder normalization, thresholding schedules, feature utilization analyses).

Important Additional Analysis:

8. No evaluation on **standard motif benchmarks (e.g., BA-Shapes, Tree-Cycles) (Shervin)** or **real-world datasets with ground-truth motifs/functional groups (e.g., benzene rings) (Harry)**, limiting external validity.
 - a. **(Shervin)** No comparison with established GNN explanation methods (**GNNExplainer/PGExplainer, CIDER, CXGNN, PAGE, SubgraphX, Attention Graphs**) to contextualize **whether SAE features offer superior fidelity or faithfulness.**
 - b. Generalization **beyond synthetic settings remains unclear; adding BA-Shapes/Tree-Cycles (Shervin) and a molecular dataset with known**

functional groups (Harry) would meaningfully increase the work's relevance.

- c. **(Shervin)** Could you compare your method's **faithfulness and ground-truth motif recovery** against **standard explainers** (e.g., GNNExplainer, CIDER, CXGNN, PAGE) on **BA-Shapes/Tree-Cycles or BA-2motif (motif benchmarks)** to contextualize the benefits of SAE features?
- d. **(Shervin)** CIDER and CXGNN explicitly pursue causal subgraph explanations with counterfactual or SCM-based objectives; PAGE uses parametric generative approaches with a causal discriminator; Attention Graphs analyze information flow in GTs/GNNs. In contrast, **this paper's focus is on feature-level mechanistic interpretability via SAEs**. While orthogonal, a **direct comparison on synthetic motif benchmarks (e.g., BA-2motif, BA-Shapes) using faithfulness metrics (fidelity/infidelity) would contextualize strengths and weaknesses.**
- e. **(Shervin)** Can you include comparisons to GNNExplainer/PGExplainer/SubgraphX on motif recovery to position the approach relative to established interpretability methods?
- f. **(Shervin)** How do the findings scale with graph size and motif density? Do monosemantic features emerge on larger graphs and under OOD motif frequency shifts?
- g. **(Harry) real-world datasets with ground-truth motifs/functional groups (e.g., benzene rings)** → If substantiated on real data, **aligning internal GNN features with interpretable graph motifs could improve trust and provide mechanistic insights in biological networks.**
 - i. Practical impact depends on demonstration beyond small synthetic graphs. **Real-world datasets (e.g., regulatory networks, citation or social networks with known motifs) and OOD settings would materially increase significance and community relevance.**
- 9. **(Harry)** Claims about **architecture-dependent monosemanticity are based on only GCN and GAT; broader generality (e.g., GraphSAGE, GIN, residual/normalized GCNs, graph transformers) is not tested.**

- a. The finding that **architectural bias affects interpretability is valuable** → could guide the design of models for domains where structure-level explanations are required. **However, the present conclusions are based on small synthetic graphs and a specific task; care is needed not to overgeneralize** claims about “fundamental trade-offs” across architectures without **broader empirical validation**.
 - b. Establishing that GNNs develop interpretable internal features aligned to graph motifs would have implications for biology, chemistry, and network science, where motif reasoning is central. **If robust, the architecture-dependent findings could guide model selection when interpretability is crucial.**
10. (Harry) Given GAT’s performance being close to the MLP baseline, do you have **diagnostics (e.g., attention distributions) indicating that GAT underutilizes the topology** in your setting? How does this affect interpretability claims?
- a. mechanistic analyses of **attention flow (Attention Graphs)**; no empirical comparisons to these families.
11. (Manhar) Statistical reporting includes extreme effect sizes (e.g., $r_{rb} \approx 1.0$) for GCN vs. MLP; while possible, more diagnostics (distribution plots, paired differences) would help verify no hidden confounds. – **See if mean baseline helps in getting better results**
- a. Some reported numbers (e.g., effect sizes of 1.00) invite deeper scrutiny and visualization. **Including code, seeds, and scripts for data generation, SAE training, and permutation testing would be important for reproducibility.**

Synthetic data questions and confounds:

12. (Shervin) Clarify possible overlap between motif labels (e.g., SIM vs. feedback): **if nodes can be in multiple motifs, the correlation structure and ablation interpretation require careful conditioning.**
13. (Shervin) Potential confounds: features **may track degree, centrality, or trivial position statistics that correlate with motif membership**. Consider partial

correlations or logistic regression controls including degree/in-degree/out-degree and edge-weight statistics to disambiguate.

- a. Did you **control for node-level confounds (degree, in/out-degree, centrality, edge-weight statistics) in the feature–motif association (e.g., via partial correlation or regression)**? If not, can you provide such controls?
14. (Shervin) How exactly are **node-level motif labels defined and assigned for overlapping motifs** within a graph? Are nodes allowed to belong to **multiple motifs** simultaneously, and how is this **handled in correlation and multiple-testing** correction?
15. How do the findings scale with graph size and motif density? Do monosemantic features emerge on larger graphs and under OOD motif frequency shifts?
16. The synthetic dataset choice (10-node graphs) enables ground-truth motif annotations, but **broader validation is needed: larger graphs, multiple motif densities, varied weight distributions, and challenging mixes**. OOD tests where **motif frequencies or compositions shift** (inspired by graph OOD literature) would **strengthen claims** about learned computational primitives.

Additions and clarifications in the paper :

17. The text presents **inconsistencies about initial state ranges ([0,1] vs. [-1,1]) and motif assignment per graph** (partition by motif type vs. “number of motif types per graph could vary”). Please **clarify the exact data generation procedure and any implications for analyses**.
18. On **which split (train/val/test)** are the point-biserial correlations computed for SAE hyperparameter selection? How do you **avoid selection bias when choosing L and k** by max $|r_{pb}|$ across many features?
19. The report of “**zero-shot generalization**” of motif-specific ablation effects to **mixed-motif graphs is anecdotal with no quantitative analysis**.
 - a. **You mention zero-shot generalization to mixed-motif graphs; could you provide quantitative results and statistical tests for those experiments?**
20. Lacking discussion and empirical comparison to post-hoc explanation methods for GNNs, motif-aware training, or causal analysis on graphs.

21. Several figures/tables are referenced but not fully shown; **key quantitative results for correlation strengths, FDR-adjusted counts of significant features, and ablation effect sizes per motif are missing.**
22. The exact **SAE training protocol (data splits for SAE, number of epochs, regularization beyond Top-K, stability across SAE seeds) is under-detailed.**
23. Some notation/typos (e.g., $z = \text{ReLU}(\text{Wench} + \text{benc})$ with unclear symbols), and figures/tables are partially truncated; key quantitative interpretability results are not visible.
24. **Overlong justifications for standard hyperparameters distract from the core scientific contributions.**

Technical limitations or concerns

The **point-biserial correlation at the node level presumes clear node-level motif labels, which is nontrivial for subgraph motifs**; labeling details and potential class imbalance handling are under-specified.

Missing related work or comparisons

The relation to motif-learning pretraining methods (e.g., MICRO-Graph) is only tangentially connected; discussion of how **SAE features differ from learned motif prototypes is missing**. → **See relevance of MICRO-graph citation and see if we wanna remove it or not**

MICRO-Graph learns **motif-like prototypes for pretraining**; discussing how SAE features differ (e.g., monosemantic latent directions vs. **prototypical subgraph embeddings**) and whether **SAE features could seed motif-guided sampling would enrich the positioning**.